

Xinyang Shen

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EDUCATION

Xidian University

BE, Software Engineering

Xi'an, China

Sep 2023 – Jun 2027

- **GPA:** 3.8/4.0 | **Comprehensive Grade:** 91.25 | **Ranking:** 27/340 (7.9%)
- **Scholarships:** University-level First-Class (2023–2024), University-level Second-Class (2024–2025)
- **Core Courses:** Computer Program Design (100/100), Compiler Principles (89/100), Computer Organization and Architecture (87/100), Discrete Mathematics (88/100)

PUBLICATION

Decoupling Confusion via Complementary Attention for Few-Shot Learning

Xinyang Shen, Shuo Li*, Fang Liu.

Pattern Recognition, Under Review

Unified Confusion Decoupling

Shuo Li, Xinyang Shen, Fang Liu*.

To be submitted

SMTrack: Subspace Memory via Low-Rank Manifold Modeling for Satellite Video Tracking

Jiahao Wang, Hao Xu, Shuo Li, Xinyang Shen, Jingyu Wang*.

ACMMM 2026 Conference

RESEARCH EXPERIENCE

Decoupling Confusion in Vision-Language Models for Few-Shot Learning

Jul 2025 – Present

Framework Lead & Core Developer (Supervisor: Dr. Shuo Li)

- Investigated the visual-semantic alignment bottlenecks of CLIP in downstream few-shot learning, identifying two intertwined challenges: inter-class confusion in the logit space and foreground-background confusion in the feature space.
- Responsible for the overall framework proposal and the complete codebase implementation, designing PDC, BSD, and CAO modules to collaboratively eliminate representation bias.
- Main author of two high-quality manuscripts, delivering a state-of-the-art 16-shot accuracy of 80.15% across 11 benchmarks; currently, one paper is under review at Pattern Recognition, and another is finalized and ready for submission.

SMTrack: Subspace Memory via Low-Rank Manifold Modeling

Nov 2025 – Mar 2026

Early-Stage Researcher (Supervisor: Dr. Jiahao Wang)

- Participated in the development of SMTrack, a satellite video tracking framework that models target appearance history as a compact low-rank manifold subspace to effectively suppress transient noise and target drift under cluttered backgrounds.
- Responsible for the preliminary literature research on satellite video vulnerabilities, implemented early-stage algorithmic verification for the anchor-guided memory read mechanism, and assisted in preprocessing the SV248S dataset.

Forest: A Privacy-Preserving Federated Learning Solution

Mar 2024 – Jul 2024

Core Developer (Supervisor: Lichuan Ma)

- Pioneered a “Cross-domain Heterogeneous Dual-engine Architecture” and a proxy-based Federated Learning scheme, fundamentally overcoming the limitations of single-modality and heterogeneous data support.
- Deployed the solution at Shenzhen Zhongtai Zhifeng Co., Ltd., achieving a recognition accuracy of 99.8% and a 27% increase in community intelligence.

System of detecting Four Treasures of Qin Mountains

Oct 2023 – Mar 2024

Project Architect & Developer

- Architected a YOLO and ESP32-CAM based real-time wildlife monitoring platform, achieving a high-precision target detection accuracy of over 95%.
- Field-tested the tracking vehicle with a 24/7 data stream, winning the First Prize (Top 1%) in the Xidian University "Spark Cup" Competition.

LEADERSHIP EXPERIENCE

Xidian University Blockchain Association

2024 – Present

Vice President

- Spearheaded the technical R&D and team cultivation within the association.
- Contributed as the 2nd author to various entrepreneurial and scientific projects, capturing multiple provincial-level accolades across high-profile business and innovation challenges.

Xidian University Student Science and Technology Association

Sep 2023 – Nov 2024

Director of the Competition Department

- Chiefly coordinated the organization of the 2024 university-level “Spark Cup” Extracurricular Academic and Scientific Works Competition, managing over 1000 project submissions and evaluation pipelines.
- Deeply involved in the overall planning, logistical arrangement, and multi-round screening process for the 2024 provincial-level “Challenge Cup” National Undergraduate Extracurricular Academic Science and Technology Contest.

AWARDS

- **Silver Award**, 15th National University Student E-commerce “ICE” Challenge 2025
- **Second Prize** in the Northwest Regional Competition, Chinese Collegiate Computing Competition 2024
- **University-level Silver Award**, China International College Students’ Innovation Competition 2024
- **Provincial-level Silver Award**, College Students’ Innovation and Entrepreneurship Training Program 2024

SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript
- **AI & Deep Learning:** PyTorch, OpenCV, Hugging Face Transformers
- **Tools & Infrastructure:** Linux, Git, Docker, SSH